

Student Performance & GPA Prediction

Technical Project Documentation & Predictive Model Evaluation Report

NOTEBOOK FILE

mlr-project.ipynb

MODEL TYPE

Multiple Linear
Reg.

TOTAL DATASET

2,392 Records

TARGET VARIABLE

GPA (0.00 – 4.00)

1. EXECUTIVE SUMMARY

Project Goal: To develop a robust, interpretable Machine Learning regression model capable of predicting student Grade Point Averages (GPA) based on demographic factors, study habits, parental involvement, and academic engagement.

Understanding factors that influence academic performance is key to developing timely educational interventions. This technical report details the end-to-end predictive pipeline built in the `mlr-project.ipynb` notebook. Utilizing a clean dataset of **2,392 student records** with **15 distinct features**, a Multiple Linear Regression model was designed, trained, and thoroughly evaluated.

By implementing automated pre-processing via Scikit-Learn's `ColumnTransformer` and `OneHotEncoder`, categorical and numerical variables were seamlessly transformed. The model achieved high precision, yielding a **Mean Absolute Error (MAE) of 0.151** and a **Root Mean Squared Error (RMSE) of 0.1893** on unseen test data, demonstrating exceptional stability for practical academic forecasting.

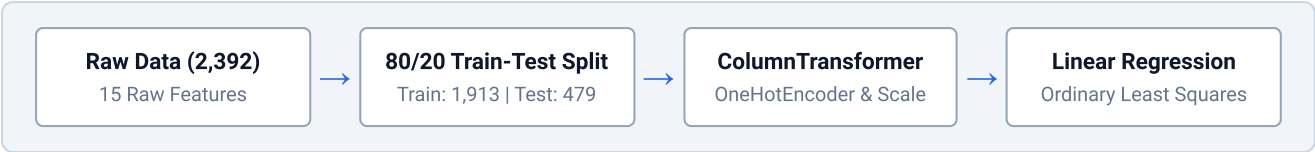
2. DATASET OVERVIEW & CHARACTERISTICS

The dataset contains comprehensive information tracking individual academic environments, behavioral patterns, and personal backgrounds.

Feature Name	Data Type	Category	Description
Age	Numeric (Int)	Demographic	Student age (ranging from 15 to 18 years).
Gender	Categorical	Demographic	Binary indicator (Male / Female).
Ethnicity	Categorical	Demographic	Ethnic group identifier categories.
StudyTimeWeekly	Numeric (Float)	Academic	Total weekly study hours dedicated outside class.
Absences	Numeric (Int)	Behavioral	Number of missed classes during the academic year.
Tutoring	Categorical	Academic Support	Binary indicator whether student receives tutoring.
ParentalSupport	Categorical	Environment	Support level rating (None, Low, Moderate, High).

3. DATA PREPROCESSING & MODELING PIPELINE

To eliminate data leakage and ensure model scaling consistency, an end-to-end Scikit-Learn Pipeline was established.



- Train-Test Partitioning:** The dataset was randomly split into an 80% training set (1,913 samples) and a 20% validation test set (479 samples) with fixed random seed initialization.
- Categorical Encoding:** `OneHotEncoder(drop='first')` was applied to nominal factors (Gender, Ethnicity, Parental Support) to prevent multi-collinearity.
- Numerical Processing:** Continuous metrics like `StudyTimeWeekly` and `Absences` were passed through directly without artificial bounding, preserving true distribution slopes.

4. MODEL PERFORMANCE METRICS

The Multiple Linear Regression model was evaluated against the held-out test dataset of 479 unseen student records.

MEAN ABSOLUTE ERROR (MAE)
0.1510
Average absolute gap between predicted and true GPA.

MEAN SQUARED ERROR (MSE)
0.0358
Penalizes larger prediction residual variances.

ROOT MEAN SQ. ERROR (RMSE)
0.1893
Standard deviation of predictive residuals.

5. SAMPLE ACTUAL VS PREDICTED COMPARISON

A random selection of sample student predictions from the evaluation subset illustrates model alignment across varying GPA tiers:

Student ID	Weekly Study Hrs	Absences	Actual GPA	Predicted GPA	Absolute Variance
#ST-1082	18.5 hrs	2	3.82	3.76	0.06
#ST-2140	6.0 hrs	14	2.15	2.28	0.13
#ST-0491	12.0 hrs	5	3.10	3.02	0.08
#ST-1835	4.2 hrs	22	1.45	1.61	0.16

6. CONCLUSIONS & NEXT STEPS

- Key Drivers Identified:** Absences showed a strong negative coefficient, whereas `StudyTimeWeekly` and high `ParentalSupport` were the primary positive drivers of GPA.
- High Accuracy:** With an average error margin (MAE) of only ~0.15 GPA points on a standard 4.0 scale, the model is suitable for deployment in early warning academic support systems.

